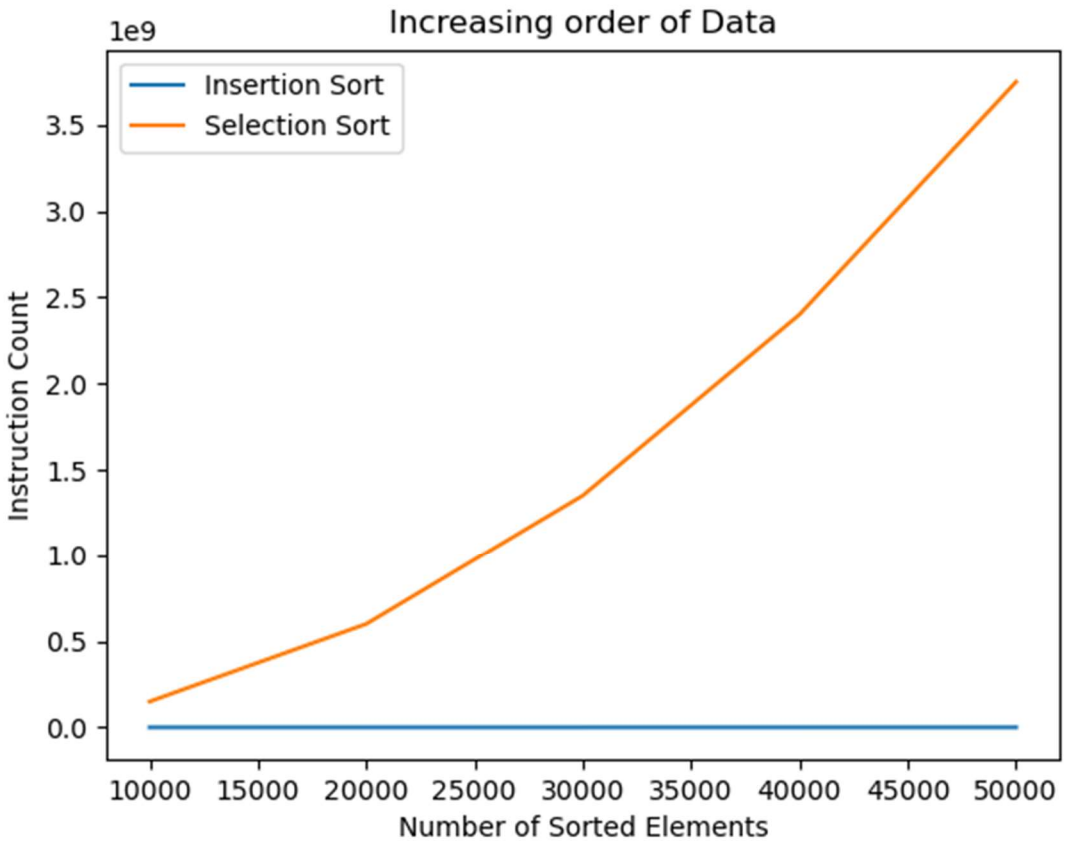

Even though the list of numbers that need to be sorted is already in an ascending order, selection sort still takes a significant number of steps. Selection sort works by checking each number in a list for the smallest value and then manually moves it into the right place. In other words, selection sort does not compare *between* elements, a comparison that would have significantly reduced the number of steps. That is also exactly why insertion sort barely takes any steps to process the list, as it is continuously making checks between values of numbers. As the list is already in the correct order, such comparisons allow it to quickly process the entire list.

Note that the number of steps of the insertion sort is not zero. It appears in such a manner on the graph as the number of steps pales in comparison to that of selection sort.